

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11: 1.2 ECOLOGY – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem? In the Kenyan river video (2:20 to 4:50), a river that once had many fish now yields far fewer, while algae and aquatic weeds grow thickly and the water sometimes smells unpleasant. Use what you learned in Lessons 1 to 9, your notes, your presentations, your nitrogen and carbon cycle drawings and your pollution summary to answer each question below. Write in full sentences in the space under each prompt. Use correct ecology terms (for example ecosystem, habitat, population, density, biotic factor, abiotic factor, carrying capacity) and give evidence from our investigations. Where a question asks for a calculation, show your working.

Part 1: The river as an ecosystem and who thrives in it

1. In Lesson 1 some learners said the fish were disappearing because they are old or have no food. Use the ecological terms you learned (ecosystem, habitat, community, biotic factors, abiotic factors) to describe the river. Then explain why 'the fish are old' or 'they have no food' cannot alone explain what you see in the video.

2. Think about the adaptations you studied in Lesson 2 (for example water hyacinth and tilapia). Explain why the conditions in the river can favour algae and aquatic weeds while harming fish.

1. The river is an ecosystem: a community of living things (fish, algae, aquatic weeds, bacteria) interacting with each other and with non-living (abiotic) factors such as water, light, oxygen and nutrients. The river is the habitat of these populations. The biotic factors are the living organisms and the abiotic factors are the non-living conditions. If the fish were just old or lacked food, we would not expect algae and weeds to boom or the water to smell, and the river is still full of life. So those ideas are incomplete. The changes must come from interactions between organisms, the environment and human activity, where a change in one part affects others.

2. Aquatic weeds such as water hyacinth are hydrophytes, adapted to living in water with air spaces and wide floating leaves, and they grow thickly when river conditions such as extra nutrients and warm water favour them. Algae also grow quickly under these conditions. Fish such as tilapia have gills and need enough dissolved oxygen in the water. When conditions change, some organisms benefit (algae and weeds) and others are harmed (fish) because each is adapted to a particular range of conditions.

Part 2: Measuring populations

3. Define population and population density. In Lesson 3

3. A population is all the individuals of one species living in an area. Population density is the number of individuals per unit area (number divided by area). Direct counting works in the school compound because the organisms (such as grass tufts or weeds)

you counted organisms in two 10 m x 10 m areas of the school compound. Explain why direct counting worked there but would be hard for the fish in the river. 4. In Lessons 4 and 5 you used quadrats and a belt transect. Describe what you found about how plants are spread in the school field and across the compound, and explain what caused the differences. 5. In Lesson 6 you marked 20 counters out of 100, then took a second sample of 20 and found 5 marked. Calculate the estimated population size and compare it with the true value. State two assumptions the method makes and explain how it could help the community find out whether fish numbers are falling.	are small in number, stay in one place and are easy to see and count in a marked area. Fish are hidden under water, move around and are too many to count one by one, so direct counting would be difficult and other methods are needed. 4. Plants are not spread evenly. Using 1 m x 1 m quadrats placed randomly, we counted very different numbers in sunny, shaded, damp and trampled spots. The belt transect across the open field, shaded area, pathway and garden showed that plant abundance changed from zone to zone as conditions changed. The differences were caused by abiotic factors (light, water, soil nutrients) and biotic or human factors (competition, grazing, trampling, gardening). Quadrat sampling lets us estimate the population without counting every plant. 5. Estimated population = (number initially marked x number in the second sample) / number recaptured = $(20 \times 20) / 5 = 400 / 5 = 80$. The true value was 100, so the estimate was close but a little lower. Assumptions: the marked and unmarked individuals mixed thoroughly; there were no births, deaths or migration during the study; marks were not lost. Repeating the trials and pooling class data gives a more reliable estimate. The community could use capture-recapture at different times and compare the estimates to see whether the fish population is falling, though the method shows how many fish there are, not why they are decreasing.
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Part 3: Factors that limit populations	
6. Use your summary of the readings and the PhET simulation from Lesson 7. Explain what biotic and abiotic limiting factors are and what carrying capacity means. Give examples of each type of factor that could affect the river fish. 7. In the discussion diamond you ranked nine factors. Choose the factor you ranked highest and explain, using evidence, how well it explains the fish decline. Explain why one factor alone is probably not enough.	6. Limiting factors are conditions that control how big a population can grow. Biotic factors are living ones such as predation, competition, disease and food. Abiotic factors are non-living ones such as temperature, oxygen, light, nutrients and water level. Carrying capacity is the maximum population size the environment can support; populations stop growing when they reach it. For the river fish, biotic examples are predators, competition for food and disease; abiotic examples are low dissolved oxygen, warm water, reduced light and extra nutrients. In the PhET simulation, adding predators made the prey population fall and then rise and fall again, and reducing food made the population decline. 7. A good answer names one factor (for example low dissolved oxygen caused by nutrient pollution, or overfishing) and gives evidence: the video shows fish decreasing while algae, weeds and smell increase, which points to nutrients and oxygen change; the simulation showed that reduced food or more predators lowers numbers. One factor alone is probably not enough because factors interact: overfishing, predators and competition act together with extra nutrients that cause algae and weeds to grow, which reduces light and dissolved oxygen.

Part 4: How nutrients move and what pollution does

8. Draw and label a diagram of the nitrogen cycle and the carbon cycle, as you did in Lesson 8, and add a pathway showing how excess nutrients enter the river. Name at least three processes in your diagram.

9. Using your pollution summary from Lesson 9, write a cause-and-effect explanation (a chain of steps) of how effluent, garbage and fertiliser runoff lead to fewer fish, thick algae and weeds, and a bad smell in the river.

10. Name two types of pollution you recorded at the shopping centre. For each, say how it could end up in a river or affect the environment. Then explain which of the pollution types you recorded or researched contribute to global warming and climate change, and why.

8. A correct drawing shows nitrogen moving between air, soil, water and living things (nitrogen fixation, nitrification, uptake by plants such as maize and beans, feeding, decomposition by bacteria) and carbon moving through photosynthesis, respiration, feeding and decomposition. Labels may include a maize shamba with beans, a dairy farm near Kericho, and sewage and farm runoff flowing to a river and Lake Victoria. An added arrow shows fertiliser, manure and sewage runoff carrying extra nutrients (nitrates and phosphates) into the river. Decomposers and bacteria are shown recycling nutrients.

9. Effluent, garbage and fertiliser runoff add nitrates and phosphates to the river, so algae and aquatic weeds grow thickly (algal bloom, eutrophication). The thick growth reduces light for plants below. When the algae and weeds die, bacteria decompose them and use up dissolved oxygen. Fish with gills do not get enough oxygen, so they suffocate or leave, and the population falls. The decomposition also produces foul-smelling gases, which explains the unpleasant smell. The river is still full of life, but the balance has changed in favour of algae, weeds and bacteria.

10. Examples: plastic bottles and polythene (solid waste) can block drains and be washed into rivers, forming garbage rivers like those we researched in the Nairobi and Athi Rivers; oily and soapy runoff from car washes flows through drains into the river and adds harmful substances. Burning waste and burning fossil fuels (smoke and exhaust fumes from matatus and boda bodas) add greenhouse gases to the air, which contribute to global warming and climate change. Plastic and oily or soapy runoff harm the river but are not the main cause of global warming.

Part 5: Restoring the river

11. Compare your Lesson 1 model of the river with your final model. Describe two ideas that changed or were added.

12. Write your Final Explanation: why are fish disappearing from the river that once supported many fish? Then propose at least three actions the community can take to restore the ecosystem, and say how each action links to a cause you identified.

11. In Lesson 1 the model showed the river with fish, algae and weeds and ideas such as old fish or lack of food. The final model includes: (a) nutrient inputs from farms, sewage, effluent and garbage entering the river; (b) the chain of algal bloom, death and decay by bacteria, loss of dissolved oxygen and smell; (c) limiting factors and interactions between biotic and abiotic factors; and (d) ways of measuring populations to monitor fish numbers. Any two of these, clearly described, are acceptable.

12. Fish are disappearing because the river is an ecosystem in which organisms and the environment interact, and human activity has changed the conditions. Effluent, garbage and fertiliser runoff enrich the river with nitrates and phosphates, so algae and aquatic weeds (hydrophytes) boom. When they die, bacteria decompose them, use up the dissolved oxygen and release foul-smelling gases. Fish, which depend on oxygen through their gills, suffocate or leave, and other factors such as overfishing, predators and competition add to the fall. Population sampling (such as capture-recapture) can show whether fish numbers are falling over time. Community actions (any three, each linked to a cause): (1) manage waste, with proper rubbish collection and unblocked drains, which reduces garbage and polythene entering the river; (2) treat effluent and reduce fertiliser and sewage runoff, which lowers nitrates and phosphates and so stops algal bloom; (3) plant riverbank vegetation to trap runoff and keep nutrients out of the water; (4) stop oily and soapy runoff from car washes from draining into the river, which removes harmful substances that damage the river. Together these can restore oxygen levels so fish populations can recover.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of ecological concepts and terms	Correctly and consistently uses ecosystem, habitat, population, density, biotic and abiotic factors, carrying capacity and adaptation to explain the river.	Uses most terms correctly with minor errors or gaps.	Uses few terms, or uses them incorrectly.
Population measurement and calculations	Accurately explains direct counting, quadrat and transect sampling, and correctly calculates the capture-recapture estimate (80) with working, assumptions and comparison to 100.	Explains the methods and calculates correctly with minor omissions in working or assumptions.	Calculation is incorrect or methods are confused or not explained.
Causal explanation of the phenomenon	Gives a complete chain: nutrient and pollution input, algal and weed growth, decay by bacteria, oxygen loss, fish death or movement, smell, plus interacting limiting factors, using evidence from the lessons.	Gives most steps of the chain with some evidence.	Gives only isolated ideas or does not link steps.
Models and use of evidence	Nutrient cycle diagram is accurate and labelled with a clear pollution pathway; refers specifically to class investigations, the video and the pollution findings.	Diagram or evidence is mostly accurate and specific.	Diagram or evidence is missing, vague or inaccurate.
Community solutions	Proposes at least three realistic Kenyan community actions, each linked to a specific cause.	Proposes two or three actions, with some links to causes.	Proposes one or no actions, or they are not linked to causes.

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Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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